Faculty of Engineering and Technology
Department of Electrical and Computer Engineering
First semester 2025/2026
Applied Cryptography-ENCS4320
HW#1

Q1) Develop a program that **analyzes multiple ciphertexts encrypted with the same 8-bit OTP key**, deduces possible plaintext characters, and demonstrates how reusing a one-time pad can compromise security.

Deliverables

Source Code (Python, C++, or Java) implementing:

- Reading multiple ciphertexts
- XOR matrix computation
- Candidate plaintext deduction
- Optional automated reduction using letter frequencies

Test Cases:

- At least **3 sets of ciphertexts** encrypted with the same key
- Show outputs for each pair: candidate plaintext lists

Q2) in this question, you will **simulate Trudy's cryptanalysis** of a Vigenère cipher.
Given:

- A known plaintext (HAPPPY)
- The corresponding ciphertext (MOGZDD)
- An upper bound on the **key length** (≤ 6)

Your task is to **determine whether the encryption key can be uniquely recovered**, and if so, to compute it **programmatically**.

Q3) in this question, you will:

1. Implement a **simple PRF**: $F_k(x)=(a\times x+b) \bmod 2^n$
2. Implement a **simple PRP** using a **Feistel Network** (with the above PRF as the round function) (3 rounds with same key)
3. Design and code a **distinguishing experiment**:
   - Compare the outputs of your PRF/PRP against a truly random function.
   - Measure how often a distinguisher can correctly guess whether they are seeing a pseudorandom or truly random mapping.

$F_k(x)=(a\times x+b) \bmod 2^n$

Where:

k = (a , b) is the key (random integers from [0, 2^n - 1])

- Input/output size = n bits (16 bits)

Q4) In this question, you will implement a program that tests whether a given encryption scheme satisfies **Shannon Perfect Secrecy**.

The program takes the following inputs:

1. **Message Space Size (m_size)**
   The message space is defined as

   $M=\{0,1,2,\ldots,m_size-1\}$.

2. **Key Space Size (k_size)**
   The key space is defined as

   $K=\{0,1,2,\ldots,k_size-1\}$.

3. **Encryption Function Enc(k, m)**
   A user-defined rule that takes a key **k** and a message **m**, and outputs a ciphertext **c**.
4. **Decryption Function Dec(k, c)**
   A user-defined rule that takes a key **k** and a ciphertext **c**, and recovers the original message **m**.

Your task is to use these definitions to evaluate whether the encryption scheme achieves **perfect secrecy**, meaning the probability distribution of messages remains unchanged even after observing the ciphertext.